

# What Can Mongolian Tell Us about Reflexivity and Reflexives?<sup>1</sup>

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## Abstract

In Mongolian, reflexivity is signaled by the clitic *-AA* instead of anaphors. Positioned at the rightmost boundary of a phrase, its function is to indicate the referential identity between a possessor in a noun phrase and a local subject or between an embedded subject and a matrix one. The reflexive strategies in Mongolian reflected by this clitic can tell us about the following. First, reflexivity is a broader cognitive property than previously defined, for example, by Reinhart and Reuland (1993). Second, reflexivity arises, in need of remedying the effect of *Inability to Distinguish Indistinguishables* (Reuland 2014), from the identity between the subjects of two predicates, the matrix one being syntactic and the embedded one being syntactic or semantic. Third, there do not exist reflexive-predicates as a grammatical category; instead, there exists a property of being “used-as-a-reflexive-predicate”. Fourth, possessives involve an abstract semantic predicate “*x HOLD y*”, where *x* is coreferential with a matrix subject in deriving possessive anaphors. Fifth, anaphors are structurally possessive DPs; they are the morphological realization of a bundle of formal features that enter the syntactic derivation of possessive structures separately.

**Keywords** reflexivity, reflexive predicate, possessive, anaphor, same-subject, Mongolian

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## Abbreviations

ACC: accusative case

ASP: aspect morpheme

DAT: dative case

GEN: genitive case

NFIN: non-finite form

NOM: nominative case

PS: passive morpheme

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<sup>1</sup> I thank XXX, XXX, and XXX for their comments on earlier versions of this paper.

PST: past tense morpheme  
RX: reflexive-possessive suffix  
∅: lack of phonological content

## 1. The current picture of reflexivity and reflexives

In *Oxford Bibliographies*, reflexivity is described by Cohen and Zribi-Hertz (2014) as a type of interpretation wherein two arguments of the same predicate co-refer. As summarized by Cohen and Zribi-Hertz (2014), scholarship on reflexivity generally follows two paths: viewing it as a specific type of anaphoric co-reference, or as one among a set of semantic effects associated with a common “reflexive” morphology. Most researchers, to my knowledge, have prioritized the former, concentrating on reflexivity in its narrowest semantic sense. Outlining the discussion in the literature of the first line of research (Faltz 1977; Reuland and Reinhart 1993; Kazenin 2001; Huang 2005; Reuland 2011, Haspelmath 2023, etc.), an informal characterization of reflexivity comes as (1) below.

- (1) Reflexivity is a dependency between two arguments of a predicate which assigns external and internal roles to them.

However, as shown below, data from Mongolian reveals that (1) cannot be maintained. Under (1), reflexivity is involved in sentences such as (2) but not those such as (3), in which the anaphor *өөр* is not an internal argument of the verb.

- (2) Baatar<sub>i</sub>      өөр<sub>i</sub>-ig-өө<sup>2</sup>      шүүмжил-sen.  
Baatar-NOM    self-ACC-RX    criticize-PST  
‘Baatar<sub>i</sub> criticized himself<sub>i/\*j</sub>.’
- (3) Baatar<sub>i</sub>      (өөр<sub>i</sub>-in)      nom-∅-oo<sup>3</sup>      mart-san.  
Baatar-NOM    self-GEN      book-ACC-RX    forget-PST  
‘Baatar<sub>i</sub> forgot his<sub>i/\*j</sub> (own) book.’

Cross-linguistically, subjects and possessive anaphors can achieve coreferentiality. Notably, this relationship does not always require explicit anaphoric marking; for instance, the Mongolian genitive anaphor *өөр-in* is optional. English similarly lacks specific anaphoric marking for subject-possessive coreference, as illustrated in (4).

- (4) John<sub>i</sub> forgot his<sub>i/j</sub> book.

At first glance, it appears that the absence of anaphors is restricted to subject-possessive coreferentiality, which is not reflexivity under (1). However, embedded anaphoric subjects are also subject to omission, as exemplified in (5).

<sup>2</sup> RX, which is subject to vowel harmony, has four allophonic variants *-aa*, *-ee*, *-oo* and *-öö*, which do not differ from each other syntactically and semantically.

<sup>3</sup> When followed by RX, the accusative marker *-ig* is not pronounced except when it is attached to the anaphor *өөр*. I use “-∅” to indicate its lack of phonological content.

- (5) Baatar<sub>i</sub> (öör<sub>i/\*j</sub>-in) buruud-san-aa meder-sen.  
 Baatar-NOM self-GEN go wrong-PST-RX admit-PST  
 ‘Baatar<sub>i</sub> admitted that he<sub>i/\*j</sub> was wrong.’

Accordingly, the absence of anaphors is true for both subject-possessive coreferentiality and subject-subject coreferentiality. Both are precluded as non-reflexive relations by the definition in (1) in that it requires that only coarguments participate in reflexivity. Reflexivity was defined as (1) under the researchers’ presupposition that anaphors are always present to mark reflexivity. Unfortunately, this is not necessarily the case. Crucially, the subject-possessive coreferentiality in (3) and subject-subject coreferentiality in (5) are marked by a dedicated suffix -*öö*, notated as RX. RX is present on the object in (3) and the embedded predicate in (5). Without RX, the book is not interpreted as belonging to the subject in (3) and the going-wrong action is not interpreted as initiated by the subject in (5).

This requires that reflexivity be defined in a broader way than (1). This said, reflexivity should be approached as a more general cognitive property than predicted by (1), with the second line of research being more necessary. Among well-known studies on narrowly defined reflexivity are Reinhart and Reuland (1993) and their subsequent works. This paper thus takes the theory of Reuland and Reinhart (henceforth, R&R) as the background of research on reflexivity in fleshing out a more general cognitive principle about it. Three core questions can be raised in approaching reflexivity.

- (6) a. What is reflexivity?  
 b. What is a reflexive predicate?  
 c. What is an anaphor?

Identifying reflexivity as a coreferential relation between coarguments, R&R formulates a reflexive predicate as follows.

- (7) A predicate is reflexive iff two of its arguments are bound by the same  $\lambda$ -operator. (Reuland 2014: 11)  
 (8) A predicate is reflexive iff one semantic argument bears two of the predicate’s semantic roles. (Reuland 2017a: 25)

These formulations render reflexivity contingent on binding, which is defined by R&R as follows.

- (9) Binding (logical-syntax based definition)  
 $\alpha$  A-binds  $\beta$  iff  $\alpha$  is the sister of a  $\lambda$ -predicate whose operator binds  $\beta$ . (Reinhart 2006: 176)

Reflexivity is thus a semantic property that is syntactically conditioned by binding and morphologically marked, with further formulations provided by R&R as follows.

(10) A predicate (formed of P) is reflexive-marked iff either P is lexically reflexive or one of P's arguments is a SELF anaphor. (R&R 1993: 678)

(11) Conditions

A: A reflexive-marked syntactic predicate is reflexive.

B: A reflexive semantic predicate is reflexive-marked. (R&R 1993: 678)

The most important ingredient of R&R's theory is Condition B in (13), which states that two identical variables on one verbal grid are avoided, and this property is expected to be universal (Reuland 2014: 12). The means to license reflexivity can be achieved by two routes including what Reuland (2014) calls *Protection* and *Reduction* (+*Bundling*), as formulated below.

(12) Protection

a. DP.  $\lambda x (V_{TR} (x, x))$

b. DP.  $\lambda x (P (x, [\text{Morph } x]))$

c. DP.  $\lambda x (P (x, f(x)))$  (Reuland 2014: 13)

(13) Reduction of an internal role

$V_{acc} (\theta_1, \theta_2) \rightarrow Rs(V) (\theta_1 - \theta_2)$

$V[\text{Agent}]_1 [\text{Theme}]_2 \rightarrow V[\text{Agent-Theme}]_1$  (Reuland 2014: 14)

R&R refutes Chomsky's (syntactic) definition of binding conditions, which are ingredients of what has been known as the Standard Binding Theory (SBT).

On the basis of a wider range of crosslinguistic data, R&R noted the shortcomings of SBT, details left out here, and argued that Principle B is strictly a condition on reflexive predicates, rather than on pronouns, regardless of their internal structure (R&R 1993: 665), and that Principle A, just like Principle B, is in fact about reflexivization, rather than about anaphors, requiring that reflexive marking be interpreted reflexively (R&R 1993: 670). The basic tenet of the R&R's theory is that a reflexive dependency is obtained between external and internal arguments by binding by a  $\lambda$ -operator. I thus refer to R&R's approach as "the  $\lambda$ -operator approach" in this paper. When I speak of Condition A/B, I refer to those of R&R and I use Principle A/B to refer to those of SBT.

Regarding the notion of anaphor, Reuland (2011: 239; 2020b: 2) holds that there does not exist an anaphor but exists an anaphor-like property. R&R distinguishes between what they call SE anaphors (simplex anaphors) and SELF anaphors (complex anaphors). SE anaphors are those that are independent from pronouns (pronominals in R&R's terms) and quite generally lack a specification for number (Reuland 2014: 6). Among the typical examples often discussed are Dutch *zich*, Icelandic *sig*, Chinese *zi-ji*, and Japanese *zi-bun*. SELF anaphors are those like English *himself*, which consists of a pronoun and an anaphor (in a looser sense). Anaphors are presumably the most important and common reflexive strategy, which exist along with reflexive clitics, verbal affixes, pronoun doubling, bodypart expressions, and putting the reflexive in a PP (Reuland 2014: 10). In some languages, a single one of these is employed, whereas in others, two or more (up to four) are employed (Reuland 2014: 6).

Being able to capture quite a wide range of crosslinguistic facts about reflexivity, R&R's

theory has developed toward a generalization, having come to us with at least two important consequences. First, it, in a principled way, brought reflexivity as a window into the way language specific principles and general cognitive principles interact (Reuland (2014: 2). Reflexivity must therefore reflect a deep property of natural language (Reuland 2014: 10). As demonstrated by Reuland (2014: 11), the definition of reflexivity in (1) provides the clue as to why reflexivity is special for the reason formulated as the *Inability to Distinguish Indistinguishables* (IDI),<sup>4</sup> which is a general property of computational systems, but not a linguistic principle. Second, reflexivity as a semantic property is manipulated by syntax. That is, binding, or more strictly, c-command and agreement, comes into play in achieving reflexivity. Reuland (2001, 2005a, 2011, 2014, 2017b, 2020a, 2020b) treats binding as an Agree operation, which involves feature (particularly phi-features) valuation between (SE) anaphors (bindees) and their antecedents (binders).

R&R’s  $\lambda$ -operator approach has focused on the following question about the specialty of reflexivity: What types of crosslinguistic variation can be envisioned? Unfortunately, R&R’s crosslinguistic research as well as others has not included Mongolian-type languages, which restricted their conception of reflexivity to (1). I believe that analyzing understudied languages like Mongolian reveals previously overlooked patterns of crosslinguistic variation. This paper demonstrates that by investigating how Mongolian licenses reflexivity, we can define it as a broader principle of Universal Grammar.

The rest of this paper is organized as follows. Sections 2 and 3 outline the basic properties of the anaphor *өөр* and RX in Mongolian. Section 4 identifies reflexivity as a coreferential relation between subjects per se, not between a subject and an object. Section 5 reasons out that the so-called “reflexive predicate” does not exist in the strict sense; what exists is a property of being “used-as-a-reflexive-predicate”. Section 6 argues that anaphors exist in human language and that they are formed by bundling a possessive feature on a D head and a reflexive feature on another into a single unit by means of “fusion”. Section 7 concludes this paper.

## 2. The anaphor *өөр* in Mongolian

In line with many other studies, this paper takes correlations between anaphors and binding as a point of departure for investigating reflexivity. Three properties of the anaphor *өөр* in Mongolian are particularly notable. First, it is not necessarily present at PF when used as a possessor and bound by a local subject (14). Second, it resists combining with a personal pronoun when used as an object (15). Third, it, whether present or absent at PF, always calls for RX (14, 15).

(14) Baatar            (өөр-in)        nom-ø-oo        mart-san.  
       Baatar-NOM   self-GEN   book-ACC-RX   forget-PST  
       ‘Baatar forgot his own book.’ (=3))

(15) \*Baatar            ter/tüün        өөр-ig-өө        шүүмжил-sen.  
       Baatar-NOM   3SG        self-ACC-RX   criticize-PST  
       ‘Intended meaning: Baatar criticized himself.’

<sup>4</sup> Thus, IDI reflects a ‘third factor’ in the sense of Chomsky (2005), as Reuland (2014) puts it.

When marked by the genitive Case ending, *өөр* can cooccur with the pronoun *түүн* but the latter is mostly absent and its presence often renders the sentence less natural to some speakers.

- (16) #Baatar (tүүн-ne) өөр-in nom-ø-oo mart-san.  
 Baatar-NOM 3SG-GEN self-GEN book-ACC-RX forget-PST  
 ‘Baatar forgot his own book.’

The genitive pronoun *түүн-ne* (or the colloquial form *тер-ne*) alone can be used in a context with a reflexive meaning, where RX must be present. However, reflexive sentences with *түүн-ne* but without *өөр-in* (17), are less natural for most speakers. In contrast, reflexive sentences with *өөр-in* but without *түүн-ne* or *тер-ne* (18), are more acceptable. Yet, they are encountered less often than those with both absent (19).

- (17) ??Baatar түүн-ne nom-ø-oo mart-san.  
 Baatar-NOM 3SG-GEN book-ACC-RX forget-PST  
 ‘Baatar forgot his book.’

- (18) Baatar өөр-in nom-ø-oo mart-san.  
 Baatar-NOM self-GEN book-ACC-RX forget-PST  
 ‘Baatar forgot his own book.’

- (19) Baatar nom-ø-oo mart-san.  
 Baatar-NOM book-ACC-RX forget-PST  
 ‘Baatar forgot his own book.’

The degree of acceptability of the related phrases is as follows.

- (20) The hierarchy of acceptability:

|              |   |               |   |                      |   |               |
|--------------|---|---------------|---|----------------------|---|---------------|
| most natural | > | natural with  | > | less natural         | > | nearly        |
|              |   | low frequency |   |                      |   | unacceptable  |
| NP-RX        |   | өөр-in NP-RX  |   | түүн-ne өөр-in NP-RX |   | түүн-ne NP-RX |

As seen from this, *өөр* shares properties with what is referred to as SE anaphors like, for example, Dutch *zich*, and is semantically equivalent to, for example, English *self*. However, *өөр* is not the only form of Mongolian anaphor. Notably, *өөр* is interchangeable with the minimal noun *bey*,<sup>5</sup> which is also interpreted as “self” in this context.

- (21) Baatar<sub>i</sub> bey<sub>i/\*j</sub>-ø-ee шүүмжил-sen.  
 Baatar-NOM body-ACC-RX criticize-PST  
 ‘Baatar criticized himself.’

<sup>5</sup> By “minimal noun”, I mean that an element such as *bey* lacks a prototypical lexical content but retains the nounhood, behaving like a lexical root. See section 6 for more details.

Interestingly, the co-occurrence of *öör* and *bey* is possible, where *öör* takes the genitive form, interpreted as the “possessor” of *bey*.

(22) Baatar<sub>i</sub>                      *öör<sub>i</sub>/\*j-in* bey-ø-ee                      *šüümjil-sen*.  
       Baatar-NOM    self-GEN-body-ACC-RX                      criticize-PST  
       ‘Baatar criticized himself.’

It is thus clear that Mongolian has all of a SE anaphor, a SELF anaphor (not productive though), and a body anaphor, the last of which contrasts with body anaphors such as *kò a li* in Haitian Creole (Rooryck and Vanden Wyngaerd 2011: 40) and *mi* in Japanese (Nishida 2002: 272; Noguchi 2018: 2) as well as head anaphors such as *kendi* in Turkish (Evseeva and Salaberri 2019: 388) and *bere burua* in Basque (Reuland 2017a: 36). However, none of these types of anaphors is the hallmark of reflexivity in Mongolian, and anaphors do not play most important roles in characterizing reflexivity in Mongolian. This suggests that anaphors might not function as true reflexivizers and that anaphoric-marked predicates may not necessarily be reflexive predicates cross-linguistically.

### 3. The reflexive clitic RX in Mongolian

As seen in section 2, the anaphor *öör* always calls for the presence of RX. As will be clear, RX is the hallmark of reflexivity in Mongolian. No reflexive interpretation is available in a sentence without RX, which occupies the rightmost position in a phrase requiring it. This means that anaphors are by no means the hallmark of reflexivity in Mongolian. This section presents empirical observations about RX, focusing on how it serves as a true reflexivizer.<sup>6</sup>

#### 3.1. Three relations marked by RX

RX, as a reflexive marker, signals certain types of dependency relations between two elements, which can be divided into three: Anaphoric relation (in the sense of R&R), possessive relation, and situational relation.

As shown in the table below,<sup>7</sup> anaphoric relations can be expressed by three different patterns, where either of *öör* and *bey* or both are used. Possessive relations are also expressed by three different patterns, where the genitive *öör-in* may or may not be used. When it is used, it may cooccur with *bey-in*, which cannot occur independently. It is noted that when *bey-in* is used, the possessive relations expressed are more restricted than the case of *öör-in*. Situational relations are expressed by only one pattern, where *öör* and *bey* are not used basically.

#### (23) Types of relations with various patterns of anaphors/NPs

| Types of relations | Patterns | Examples |
|--------------------|----------|----------|
|--------------------|----------|----------|

<sup>6</sup> Explanatory discussions on RX can be found in studies including Anisman (2010), Guntsetseg (2011, 2012), Maki et al. (2015), Kazama (2019), Bai and Cao (2024) and Gong (2023, 2024), and descriptive discussions can be found in Poppe (1954), Street (1963), Janhunen (2012), Kullman and Tserenpil (2015) in addition to local grammar books written in Mongolian.

<sup>7</sup> K represents a Case marker, and RX represents the reflexive marker. Note that the accusative Case marker *-ig* is optionally absent and is so especially in colloquial Mongolian.



|             |                                                                    |                                                                                    |
|-------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------|
| anaphoric   | <i>öör</i> -K-RX;<br><i>bey</i> -K-RX;<br><i>öör-in bey</i> -K-RX; | <i>öör-ig-öö</i> ;<br><i>bey-ø-öö</i> ;<br><i>öör-in bey-ø-ee</i> ;                |
| possessive  | NP-K-RX;<br><i>öör-in</i> NP-K-RX<br><i>öör-in bey-in</i> NP-K-RX  | <i>nom-ø-oo</i> ;<br><i>öör-in nom-ø-oo</i> ;<br><i>öör-in bey-in čadal-ø-aa</i> ; |
| situational | NP-K-RX                                                            | <i>nom-oo</i>                                                                      |

Anaphoricity (or anaphoric relation) here corresponds with the reflexivity (or reflexive relation) characterized by R&R, who excludes possessive reflexivity from reflexivity, as shown in (24). Therefore, according to R&R, sentences such as *John<sub>i</sub> loves his<sub>i</sub> dogs* do not express reflexivity, although coreferentiality is obtained between the subject and the possessive pronoun. Situational reflexivity is not attested in the languages studied by R&R, and no relevant discussion has been made. Situational reflexivity will be shown to be a subtype of reflexivity in the remainder of this paper.

(24) Types of reflexivity in this paper and R&R's:

| This paper                       | R&R                           |
|----------------------------------|-------------------------------|
| anaphoric relation/reflexivity   | identified as reflexivity     |
| possessive relation/reflexivity  | not identified as reflexivity |
| situational relation/reflexivity | not discussed                 |

An anaphoric relation is obtained between a subject and an object, where the latter must be anaphoric-marked, hence the name “anaphoric relation”. In signaling an anaphoric relation in Mongolian, RX must cooccur with an anaphor, *öör* or *bey*, which takes accusative Case, as exemplified in (21), (22) and (25) below.

- (25) Baatar<sub>i</sub>            *öör<sub>i</sub>/\*j-ig-öö*            *šüümjil-sen*.  
Baatar-NOM   self-ACC-RX   criticize-PST  
‘Baatar criticized himself.’

A possessive relation is obtained between a subject and a possessor in an object DP, the latter of which may or may be anaphoric-marked, hence it is not an anaphoric relation. In Mongolian, when a possessive relation is obtained, the absence of *öör-in* is optional, as exemplified in (26).

- (26) Baatar            (*öör-in*)            *nom-ø-oo*            *mart-san*.  
Baatar-NOM   self-GEN   book-ACC-RX   forget-PST  
‘Baatar forgot his own book.’ (= (3/14))

A situational relation is obtained between a subject and an element which is neither an object nor a possessor. That element may be either a nominal or a predicate. For RX to signal a situational relation involving a nominal, the determiner *nögöö*, which literally translates as “the other”, is preferred instead of the genitive anaphor *öör-in*, as exemplified in (27). *Nögöö* here,



which often comes with a relative-clause nuance,<sup>8</sup> indicates that the book has appeared in the discourse and is an old information. *Nөгөө nom* was translated as “the book which we talked about recently” by Guntsetseg (2011). However, *нөгөө* does not necessitate the book’s connection with a prior event. It is not necessarily the case that an actual conversation about the book took place prior to the speech time. *Nөгөө* instead entails that the referent of the noun is engaged in two eventualities including the proposition uttered in the sentence and the one in a presupposition. When cooccurring with RX, *нөгөө* requires that the initiator of the presupposed eventuality be the same as that of the proposed eventuality. Therefore, a more appropriate interpretation of (27) would be “Baatar has read the book which he is obligated to read”.

- (27) Baatar            нөгөө    nom-ø-oo            унš-san.  
       Baatar-NOM    that        book-ACC-RX    read-PST  
       ‘John read the book (which we talked about recently).’ (adapted from Guntsetseg 2011)

Let us refer to such relative clauses as “implicit relative clauses”, just for simplicity purposes. Implicit relative clauses can have overt counterparts, where the relative subject must be coreferential with the matrix subject.

A situational relation is also obtained in overt relative clauses such as (28) in the same manner. In (28), the lesson is not possessed by the subject but has a relation with it, which is imposed by the situation in which the subject performs a learning action on the lesson.

- (28) Baatar            sur-san        hičeel-ø-ee            mart-san.  
       Baatar-NOM    learn-PST    lesson-ACC-RX    forget-PST  
       ‘Baatar forgot the lesson that he learned.’

The function of RX here is to indicate that the lesson is already related to the subject in the embedded proposition when it occurs in the matrix proposition, where the proposed eventualities share the same subject. In other words, RX functions to indicate the same-subject of the two predicates.

A situational relation can also be established between an action and its performer. That is, RX, when attached to a clausal argument, signals a situational relation between the subordinate clause and the matrix subject. In (29), the subject performs an asking-for-leave action, where a relation naturally arises between the subject and the action. Notably, the action enters into a relation with the subject for the second time by virtue of participating in the forgetting event performed by the subject. RX is present there to indicate that the asking-for-leave action as a participant of the forgetting action has a situational relation with the subject and that the actions share the same subject.

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<sup>8</sup> When used for expressing situational relations, *нөгөө* is functionally equivalent to the “the ... that the subject ...” configuration in English. For example, *нөгөө nom* in (27) is most likely used in a context in which, for example, I have a duty of doing something about a book and I speak of it in an utterance. In this sense, the hidden relative clause denotes a “duty” event in the speaker’s presupposition. This property of *нөгөө* makes it very subject to Kayne’s (1994) analysis of relative clauses. *Nөгөө* can also be used in its prototypical meaning, without expressing a situational relation. In that case, *нөгөө nom* means “the other book”, rather than “the ... that the subject ...”, with no relative-clause nuance available.

- (29) Baatar            cölöö guya-h<sup>9</sup>-aa            mart-san.  
       Baatar-NOM   ask for leave-NFIN-RX   forget-PST  
       ‘Baatar forgot to ask for leave.’

What is then the condition or principle underlying these three relations, which are all marked by the uniform reflexivizer RX? This is one of our primary concerns in the remainder of this paper. A detailed discussion is provided in sections 4 and 5. To that end, we proceed to scrutinize the same-subject property related to the reflexive semantics of RX.

### 3.2. Same-subject as the underlying relation

In a nutshell, the principle underlying the three semantic relations is that RX occurs only where two subjects are coreferential. As noted above, the possessive anaphor *öör-in* always calls for the presence of RX on the possessed NP as in *öör-in nom-oo* ‘one’s own book’ in a non-nominative position, as in (31). Importantly, *öör-in* itself can host RX, as exemplified, as in (30). That is, the host of RX is optional; it is either the genitive anaphor *öör-in* or the NP. However, *öör-in* is not obligatory at PF and when it is absent, RX is attached to the possessum, as exemplified by (32).

- (30) Baatar            öör-in-öö            nom-ig            mart-san.  
       Baatar-NOM   self-GEN-RX   book-ACC   forget-PST  
       ‘Baatar forgot his (own) book.’

- (31) Baatar            öör-in            nom-ø-oo            mart-san.  
       Baatar-NOM   self-GEN   book-ACC-RX   forget-PST  
       ‘Baatar forgot his (own) book.’

- (32) Baatar            nom-ø-oo            mart-san.  
       Baatar-NOM   book-ACC-RX   forget-PST  
       ‘Baatar forgot his (own) book.’

Two RXs, each for *öör*, which is the possessor, and the object, which is the possessum, are not allowed.

- (33) \*Baatar            öör-in-öö            nom-ø-oo            mart-san.  
       Baatar-NOM   self-GEN-RX   book-ACC-RX   forget-PST  
       ‘Intended meaning: Baatar forgot his (own) book.’

RX is disallowed for an object that is not “possessed” by the subject. Therefore, (34) will be out under the reading that Baatar forgot someone else’s book.<sup>10</sup>

<sup>9</sup> Nominalized complement clauses are not accusative-marked when RX is present.

<sup>10</sup> “Possession” in question does not necessarily refer to a prototypical relationship where a possessor exerts actual physical control or ownership over an object. It always extends to include a dealing-with relation. For example, in (19), when the possessive pronoun *öör-in* is absent, the book is either actually possessed by the

- (34) \*Baatar            nom- $\emptyset$ -oo            mart-san.  
       Baatar-NOM    book-ACC-RX    forget-PST  
       ‘Intended meaning: Baatar forgot someone else’s book.’

Unlike the possessive anaphor *öör-in*, which can be separated from RX, a possessive NP must not be separated from RX. In (35), *bagš* ‘teacher’ denotes the subject’s teacher. However, RX is not attached to *bagš* but rather to *nom* ‘book’, which is not possessed by the subject, which leads to the ungrammaticality of the sentence.

- (35) \*Baatar            bagš-in            nom- $\emptyset$ -oo            mart-san.  
       Baatar-NOM    teacher-GEN    book-ACC-RX    forget-PST  
       ‘Intended meaning: Baatar forgot his teacher’s book.’

RX can be attached to *bagš* when the teacher is Baatar’s teacher.

- (36) Baatar            bagš-in-aa            nom-ig            mart-san.  
       Baatar-NOM    teacher-GEN-RX    book-ACC    forget-PST  
       ‘Baatar forgot his teacher’s book.’

Again, two instances of RX, for the possessor and the possessum, are not allowed.

- (37) \*Baatar            bagš-in-aa            nom- $\emptyset$ -oo            mart-san.  
       Baatar-NOM    teacher-GEN-RX    book-ACC-RX    forget-PST  
       ‘Intended meaning: Baatar forgot his teacher’s book.’

In summary, in the context in which the subject NP2 is coreferential with the possessor of NP1, the possessor must be realized or interpreted as the pronoun *öör-in*, and the host of RX varies between *öör-in* and NP1. If NP2 is not coreferential with NP3, which is the possessor of NP1, then NP3 is necessarily the host of RX.

(38) Distributional paradigm of RX in object NPs<sup>11</sup>

|   | NOM              | GEN                        | ACC                     | Example |
|---|------------------|----------------------------|-------------------------|---------|
|   | NP2 <sub>i</sub> | <i>öör<sub>i</sub></i> -RX | NP1 <sub>j</sub>        | (30)    |
|   | NP2 <sub>i</sub> | <i>öör<sub>i</sub></i>     | NP1 <sub>j</sub> -RX    | (31)    |
|   | NP2 <sub>i</sub> |                            | NP1 <sub>j</sub> -RX    | (32)    |
| * | NP2 <sub>i</sub> | <i>öör<sub>i</sub></i> -RX | NP1 <sub>RELj</sub> -RX | (33)    |
| * | NP2 <sub>i</sub> | NP3 <sub>k</sub> -RX       | NP1 <sub>RELj</sub> -RX | (37)    |
|   | NP2 <sub>i</sub> | NP3 <sub>k</sub> -RX       | NP1 <sub>j</sub>        | (36)    |
| * | NP2 <sub>i</sub> | NP3 <sub>k</sub>           | NP1 <sub>j</sub> -RX    | (35)    |

subject or acted on (that is, dealt with) by him, or both. See also Guntsetseg (2012: ch.3) for discussion of RX in this use.

<sup>11</sup> Case suffixes are all left out here, (46), and (54).

When RX is attached to a noun that heads an NP containing an object relative clause, the subject of the relative clause must be the genitive anaphor *өөр-in*, which is coreferential with the matrix subject. The relative subject *өөр-in* may be either absent as in (39) or present as in (40).

- (39) Baatar            sur-san            hičeel- $\phi$ -ee            mart-san.  
       Baatar-NOM   learn-PST   lesson-ACC-RX   forget-PST  
       ‘Baatar forgot the lesson that he learned.’

The non-genitive anaphor *өөр* is out. Note that in Mongolian, the subject of object relative clauses must be genitive, regardless of whether it is coreferential with the matrix subject. With *өөр-in* present, the sentence sounds less natural to some speakers. RX is present on either *өөр-in* or the noun. The co-occurrence of two instances of RX are not allowed. The examples in (40–42) illustrate this.

- (40) Baatar            өөр-in            sur-san            hičeel- $\phi$ -ee            mart-san.  
       Baatar-NOM   self-GEN learn-PST   lesson-ACC-RX   forget-PST  
       ‘Baatar forgot the lesson that he learned.’

- (41) Baatar            өөр-in-өө            sur-san            hičeel-ig            mart-san.  
       Baatar-NOM   self-GEN-RX   learn-PST   lesson-ACC   forget-PST  
       ‘Baatar forgot the lesson that he learned.’

- (42) \*Baatar            өөр-in-өө            sur-san            hičeel- $\phi$ -ee            mart-san.  
       Baatar-NOM   self-GEN-RX   learn-PST   lesson-ACC-RX   forget-PST  
       ‘Intended meaning: Baatar forgot the lesson that he learned.’

If the relative subject is not coreferential with the matrix subject but possessed by it, then RX is attached to it.

- (43) Baatar            bagš-in-aa            zaa-san            hičeel-ig            mart-san.  
       Baatar-NOM   teacher-GEN-RX   teach-NPST   lesson-ACC   forget-PST  
       ‘Baatar forgot the lesson that his teacher taught.’

RX is out when switch reference (SR) is obtained between the relative subject and the matrix subject.

- (44) \*Baatar            bagš-in            zaa-san            hičeel- $\phi$ -ee            mart-san.  
       Baatar-NOM   teacher-GEN   teach-NPST   lesson-ACC-RX   forget-PST  
       ‘Intended meaning: Baatar forgot the lesson that his teacher taught.’

For a subject relative clause, RX is present on the object, which is possessed by the subject, as exemplified below.

- (45) Baatar            gee-gd-sen    nom-ø-oo            olž ab-san.  
       Baatar-NOM   lose-PS-PST   book-ACC-RX   find-PST  
       ‘Baatar has found his book that was lost.’

In summary, we obtain the following paradigm for object NPs containing (object) relative clauses, notated as NP<sub>REL</sub>.

(46) Distributional paradigm of RX in object NPs with (object) relative clauses

|   | NOM              | GEN                  | ACC                     | Example |
|---|------------------|----------------------|-------------------------|---------|
|   | NP2 <sub>i</sub> | öör <sub>i</sub> -RX | NP1 <sub>RELj</sub>     | (41)    |
|   | NP2 <sub>i</sub> | öör <sub>i</sub>     | NP1 <sub>RELj</sub> -RX | (40)    |
|   | NP2 <sub>i</sub> |                      | NP1 <sub>RELj</sub> -RX | (39)    |
| * | NP2 <sub>i</sub> | öör <sub>i</sub> -RX | NP1 <sub>RELj</sub> -RX | (42)    |
|   | NP2 <sub>i</sub> | NP3 <sub>k</sub> -RX | NP1 <sub>RELj</sub>     | (43)    |
| * | NP2 <sub>i</sub> | NP3 <sub>k</sub>     | NP1 <sub>RELj</sub> -RX | (44)    |

Turning to the case in which RX is attached to a nonfinite verb signaling the boundary of an object clause, the embedded subject, mostly implicit, must refer back to the matrix subject.

- (47) Baatar            buruud-san-aa    meder-sen.  
       Baatar-NOM   go wrong-PST-RX   admit-PST  
       ‘Baatar admitted that he was wrong.’

However, the presence of the embedded subject in the form of *öör-in* makes the sentence sounds a bit less natural to some speakers.

- (48) Baatar            öör-in            buruud-san-aa    meder-sen.  
       Baatar-NOM   self-GEN   go wrong-PST-RX   admit-PST  
       ‘Baatar admitted that he was wrong.’

When RX is present on *öör-in*, the sentence still sounds less natural than the case with *öör-in* absent but not unacceptable. The lexical semantics and (non)finiteness of the verb and pragmatics may affect the acceptability.

- (49) Baatar            öör-in-öö            buruud-san-ig    meder-sen.  
       Baatar-NOM self-GEN-RX   go wrong-PST-RX   admit-PST  
       ‘Baatar admitted that he was wrong.’

- (50) #Baatar            öör-in-öö            unta-h-ig            mart-san.  
       Baatar-NOM   self-GEN-RX   sleep-NFIN-ACC   forget-PST  
       ‘Baatar forgot to sleep.’

RX is disallowed to be present on *öör-in*, when RX is present on the verb.

- (51) \*Baatar            öör-in-öö            buruud-san-aa            meder-sen.  
 Baatar-NOM    self-GEN            go wrong-PST-RX    admit-PST  
 ‘Intended meaning: Baatar admitted that he was wrong.’

RX is out when SR of subjects is obtained, as shown below.

- (52) \*Baatar            Bat-in            buruud-san-aa            med-sen.  
 Baatar-NOM    self-GEN    go wrong-PST-RX    know-PST  
 ‘Intended meaning: Baatar realized that Bat was wrong.’

RX can be present on the genitive subject in the object clause when it is possessed by the matrix subject, as shown below.

- (53) Baatar            bagš-in-aa            buruud-san-ig            med-sen.  
 Baatar-NOM    teacher-GEN-RX    go wrong-PST-ACC    know-PST  
 ‘Intended meaning: Baatar realized that his (=Baatar) teacher was wrong.’

Summing up, we obtain a similar distributional paradigm of RX for the “nominalized” nonfinite verb heading an object clause.

(54) Distributional paradigm of RX in object clauses

|   | NOM              | GEN                  | ACC  | Example |
|---|------------------|----------------------|------|---------|
|   | NP1 <sub>i</sub> | ööri-RX              | V    | (49)    |
|   | NP1 <sub>i</sub> | ööri                 | V-RX | (48)    |
|   | NP1 <sub>i</sub> |                      | V-RX | (47)    |
| * | NP1 <sub>i</sub> | ööri-RX              | V-RX | (51)    |
|   | NP1 <sub>i</sub> | NP2 <sub>j</sub> -RX | V    | (53)    |
| * | NP1 <sub>i</sub> | NP2 <sub>j</sub>     | V-RX | (52)    |

In conclusion, RX signals a dependency between subjects and non-nominative elements, which are either embedded subjects or possessors in object DPs, and is incompatible with SR. RX is always attached to the possessum when the possessor is an NP subject, whether it is of matrix clauses or of object clauses. It is attached to the possessor only when the possessor is the genitive *ööri-in* that is coreferential with a subject.

A few remarks on SR are in order. Striling (1993: 6), on the basis of investigating North American languages and others, states that SR holds between just two clauses that are structurally local and linearly adjacent. SR markers often occur in the dependent clause. However, markers for conjoint reference of subjects are also observed, as opposed to SR markers. In either case, there will be a marker that distinguishes different subjects or identifies one as another, avoiding confusion or misunderstanding. This is exactly what *the Inability to Distinguish Indistinguishables* as a general property of computational systems (Reuland 2014) predicts. XXX observes that Mongolian is one of the languages taking the option of marking distinctness, not identity, of subjects. He views that the differential-subject marking property and the reflexive-possessive property of Mongolian represent two separate grammatical

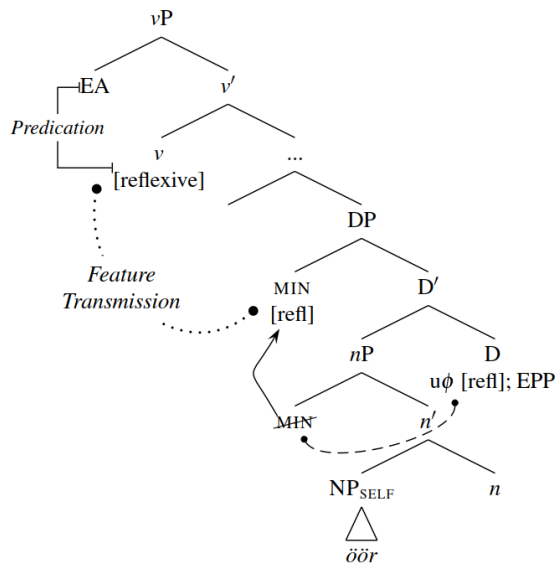
systems of SR in the language. To me, however, Mongolian takes both options; that is, Mongolian has a differential-subject marking property (Guntsetseg and Klein 2009; von Heusinger et al. 2011; Guntsetseg 2012; Hsiao 2012), a SR property, for marking disjoint reference of subjects, and RX for marking conjoint reference of subjects, both of which have the same effect of avoiding confusion. The properties of RX summarized in (38) suggest that marking conjoint reference is not limited to clauses; it is required between a clause and a possessive DP, in which the possessor acts as an embedded subject in one way or another. This is what RX contributes in the grammatical system concerned.

### 3.3. Categorical status of RX

It is quite clear from the above discussion that in Mongolian it is RX rather than the anaphor *өөр* that is the reflexive marker. Note that *өөр* can be absent while RX is always required in a reflexive context.

With respect to the categorial status of RX, traditional grammarians refer to it as a suffix, without discussing its exact status. Among theoretical studies, Maki et al. (2015: 67ff) labeled RX “pronoun” and Gong (2023) treated it as a D head. Maki et al. (2015: 67ff) proposed that RX undergoes LF movement to its antecedent, e.g., the subject. However, RX can never be an element that can be characterized as a reflexive pronoun. It lacks a specification of any nominal features, which a pronoun may have. Functionally, RX is similar to anaphors (or reflexive pronouns), but it should belong to a different category. According to Gong (2023), RX as a D head selects *nP*, where *өөр*, she assumes, is a noun much like a common noun.

(55) Nature of RX: Gong (2023: 5)



Gong (2023) made the following specific proposals.

(56) RX (her REFL.POSS -AA) signals agreement with a minimal pronoun bound by the closest *v* which introduces the local subject.

- i. Binding mechanism: Binding by *v*.
- ii. Nature of bound (reflexive) pronoun: Minimal Pronouns (MIN)



### iii. Morpho-syntactic source of -RX: Feature Transmission

Gong's (2023) proposal follows from Kratzer's (2009), who argues that it is the local functional head  $\nu$  rather than an antecedent DP that serves as the binder of reflexives, which acquire their phi-feature set from  $\nu$  via Feature Transmission under Binding (FTUB).

#### (57) Feature Transmission under Binding (Kratzer 2009: 216)

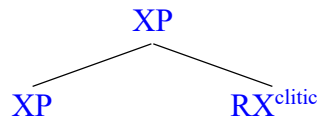
The  $\phi$ -feature set of a locally bound pronoun unifies with the  $\phi$ -feature set of the head that hosts its binder.

On this account, reflexives are elements born without  $\phi$ -features, which Kratzer (2009) called MIN, for minimal pronoun. Following Kratzer (2009), Gong (2023) assumes that there is MIN in Mongolian, which is base-generated in Spec of  $nP$ , sharing  $\phi$ -features with D, and moves to Spec of DP due to EPP on D. Having moved to Spec of DP, MIN remains in the same domain as  $\nu$ . FTUB applies and MIN receives the feature [ref], for reflexivity, from  $\nu$ . Since MIN shares this feature with D, as a result it is present on three heads including  $\nu$ , MIN, and D, and is spelled out as RX on D.

However, assuming RX as D fails to account for the fact that RX is always the outermost element within a DP; it follows any other nominal suffixes including those for number and case. Assuming it as a feature receiver is also problematic with FTUB, which itself, as discussed by Reuland (2020a: 5), is an inappropriate assumption given that feature transmission is not feature valuation but rather a feature checking.

Given that RX is an outermost element in a DP, it is supposed to occupy an adjunct/specifier position. However, this would mean that it is a phrasal element, contra the fact that it does not display any properties displayed by phrases; it lacks a specification of phi-features (person, gender and number) and is unable to move. This in turn indicates that it is a head-like element.<sup>12</sup> This ambivalent property of RX suggests that it in fact is a clitic-like element. Note that clitics are generated as specifiers of a null head (Bošković 2002). This leads us to the following representation, in which RX sits in the rightmost specifier of DP.<sup>13</sup>

#### (58) RX as a clitic:



The status of RX as a reflexivizer presents it as a novel case in which reflexivity can also be licensed by a nominal clitic. Reflexivity with RX does not seem to belong to any of the five categories of reflexives presented by Dechaine and Wiltschko (2017). Dechaine and Wiltschko (2017: 64ff) argued that five well-established positions including D, phi, Class, n, or N are available in the extended projection of the nominal phrase, each being associated with a reflexive form. Note that RX is even above D, as indicated by the data discussed above. RX

<sup>12</sup> See Chomsky (1995) and Bošković (2002) for relevant discussion on the ambiguous property of clitics, which behave like both  $X^0$  or XP.

<sup>13</sup> See Bai (2025; manuscript) for a detailed discussion on the categorial status and position of RX.

arguably occupies a position closer to the head of KP (Kase phrase). This status of RX brings to us a question of what it can mean for the theory of reflexivity.

#### 4. What is reflexivity?

Elaborating on the same-subject property discussed in section 3.2, this section works out a novel definition of reflexivity. In a nutshell, this section justifies (59) and (60) rather than (61).

(59) Reflexivity arises between specifiers, not between a specifier and a complement.

(60) Reflexivity is a dependency between the external argument of a syntactic predicate and that of a semantic one.

(61) Reflexivity is a dependency between two arguments of a predicate which assigns external and internal roles to them. (=1))

The crux of matter is to explicate the property of anaphors as possessive DPs. Crucially, the Mongolian facts examined in sections 2 and 3 as well as those about reflexivity reported in the literature strongly suggest that anaphors are spell-out forms of a possessive DP structure. That is, an anaphoric relation is in fact a subcase of a possessive relation.

To elaborate on this issue, recall that in Mongolian *bey* ‘body’ can be used the same way as the anaphor *ööṛ*.

(62) Baatar<sub>i</sub>            öör<sub>i</sub>/<sub>\*</sub>j-ig-öö       šüümjil-sen.  
Baatar-NOM   self-ACC-RX   criticize-PST  
'Baatar criticized himself.'  
(= (25))

(63) Baatar<sub>i</sub>            bey<sub>i</sub>/\*j-ee            šüümjil-sen.  
Baatar-NOM   body-ACC-RX   criticize-PST  
'Baatar criticized himself.'                                 (= (21))

(64) Baatar<sub>i</sub>            öör<sub>i</sub>/\*<sub>j</sub>-in bey-ee            šüümjil-sen.  
Baatar-NOM    self-GEN-body-ACC-RX    criticize-PST  
'Baatar criticized himself.'  
(= (22))

The contrast between the last two sentences shows that *bey* in this use is logically a possessum, possessed by *öör*, which behaves like a possessor, referring back to the subject. In this sense, *bey* is the lexical core and *öör-in* is the possessive determiner within the DP structure as represented below.

(65) [<sub>XP</sub> ... subject ... [<sub>DP</sub> possessor [<sub>D'</sub> D [<sub>N</sub>]]]

(66) [<sub>XP</sub> ... Baatar ... [<sub>DP</sub> *öör* [<sub>D'</sub> -in [*bey*]]]

This is supported by and accounts for the fact that languages allow bodypart anaphors literally

meaning “self’s body”.

Assuming that the structure in (65) holds universally, what is selected by D, a possessive determiner head, is either a noun (phrase) with maximal lexical content or a root-like noun with minimal lexical content. In any case, there is a lexical core selected by D. The way of lexicalizing DP, which is not universal, then varies among languages. One language may allow a single morphological unit to spell out any two or all of the D head, its specifier and its complement N, while another may allow respective morphemes for the three elements. Due to economy principles about spelling out,<sup>14</sup> it is expected that redundancy of morphology is diminished so that any of the three can and even must be absent from PF.

Conceptually, a possessive-possessum relation is a kind of subject-object relation mediated by a functor. In this regard, the predicate is something like “*x holds y*”, where *holds* is a possessive predicate, *x* is an external argument,<sup>15</sup> and *y* is an internal argument. Roversi (2024) observes that an abstract transitive verb, what he calls POSS, takes the possessor as its external argument and the possessum as its internal argument, which gives rise to the same possessive semantics as the nominal head *ʒ*.

(66) [v<sub>P</sub> possessor [v' POSS [DP possessum]]] (Roversi 2024: 386)

Roversi (2024) argues that in languages such as Äiwoo possessive DPs are derived by relativization of the possessum, as shown below.

(67) Derivation of *my boat*:

[DP boat<sub>i</sub> [RC ... [v<sub>P</sub> I [v' POSS [DP \_\_\_\_<sub>i</sub> ]]]]] (adapted from Roversi 2024: 386–393)

This derivation, according to Roversi (2024), is not supposed to be universal. However, it provides us with a way into the logical structure of possessums. Based on (66), we paraphrase (68) as (69).

(68) Pasha loves her cat.

(69) Pasha loves the cat she holds/possesses/has.

(70) [X<sub>P</sub> Pashaj [v<sub>P</sub> love [DP cat<sub>i</sub> [RC [v<sub>P</sub> she<sub>j</sub> [v' \_\_\_\_<sub>i</sub> ]]]]]]]

Note that we are not trying to derive possessive DPs by relativization. We are instead

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<sup>14</sup> A principle of economy under discussion can be stated as follows.

(i) Use as few morphemes as possible, where other principles, if any, leave the choice open.

See also (84) and relevant discussion in section 5.

<sup>15</sup> R&R (1993) and many others also treat the possessor in a possessive DP as an external argument/subject. By this, R&R (1993: 682) explained the difference in grammaticality between the following sentences. In (ib), *your* is present as the subject of a semantic predicate and therefore subject to Condition B in (11), leading to the ungrammaticality, with no coreferentiality obtained between it and *herself*.

(i) a. Lucie liked [(a) picture of herself].  
b. \*/?Lucie liked [your picture of herself].

representing the logical syntax of them.<sup>16</sup> Given this, reflexivity arises between the subject and the possessor, which is “the subject of an implicit relative clause”, not between the possessum and the subject. Alternatively speaking, reflexivity arises from the referential identity between external arguments, one being a subject and another being a possessor or another subject. Given that subjects and possessors are normally located in specifiers, we obtain (59). This goes contrary to R&R’s definition, which requires that reflexivity arises between external and internal arguments.

An anaphoric relation, a possessive relation, and a situational relation are thus brought under a unified frame. Recall that a situational relation holds between two subjects, one being matrix another being embedded. Note that the embedded subject may be one of an abstract (implicit) relative clause, as indicated by our description of the *nogöo* example in section 3.1.

Given that possessors are subjects of *HOLD*, it is a same-subject relation that is the source of reflexivity. Therefore, the three relations signaled by RX are distinct spell-outs of a same-subject relation.

Note that when I speak of same-subject, “subject” means either the subject of uttered clauses or that of abstract relative clauses in possessive DPs including radical possessives such as *my dog* and anaphoric possessives such as *my-self* and *öör-in bey* ‘self’s body’. The intimacy between possessive DPs and embedded clauses is also evidenced by genitive subject constructions. In English, nonfinite clauses can be nominalized, taking the *ing* form, with the subject, when present, being genitive. As observed by studies such as Abney (1987) and others, the morpheme *ing* spells out a nominalizing head that selects the verbal core as its complement to form an NP and that the genitive subject occupies the specifier of a D head selecting the NP. In this sense, the genitive subject is some kind of possessor that holds the event as a possessum under its control.

(71) I learned about John’s smoking stogies. (Abney 1987: 109)

In Mongolian, the subject of relative clauses obligatorily takes genitive Case and that of complement clauses optionally takes it.<sup>17</sup>

(72) Baatar            bagš-in            zaa-san            hičeel-ig            mart-san.  
       Baatar-NOM    teacher-GEN    teach-NPST    lesson-ACC    forget-PST  
       ‘Baatar forgot the lesson that the teacher taught.’

(73) Baatar            Bat-in            cölöö guya-h-ig            med-ne.  
       Baatar-NOM    Bat-GEN    ask for leave-NFIN-RX    know-NPST  
       ‘Baatar knows Bat’s asking for leave.’

<sup>16</sup> Logical syntax is a formal representation of the output of the computational system with the degree of detail required by the inference system (and independently justified because of the requirements of the inference system); essentially, it is syntax with an extended vocabulary (Reuland 2011: 34). It is a representation of linguistic structure that is sufficiently fine grained to feed the inference system (Reuland 2014: 28).

<sup>17</sup> See von Heusinger et al. (2011) for related discussion.

Notably that genitive subjects are disallowed in clauses headed by *gež*, which acts as a complementizer (Gong 2022: ch.2), and that the verb endings *-san* and *-h* (as well as *-dag*), which are called adjectival verb suffixes in Mongolian grammar (Cinggeltei 1997: 266–268), signify the deficiency of C-properties. *-San* is special in that it expresses both past tense and perfective aspect and that it acts as an adjectivizing head, as in *hel-sen üg* ‘spoken word’.<sup>18</sup> Note also that *-san* and *-h* can be attached by nominal Case markers. These findings suggest that genitive subjects in Mongolian have to do with the noun-like status of their predicates,<sup>19</sup> which are not CPs. Kornfilt and Whitman (2011; 2012) observed that TPs (in Japanese and Turkish) are subject to nominalization, taking genitive subjects. Thus, the linkage between genitive subjects and nominalization of non-CP clauses is by no means arbitrary. It reflects a cognitive property that events, similar to entities, are “held” or “controlled” by their subjects. Thus, genitive subjects and nominalization of non-CP clauses make the subject-predicate relation resemble the possessor-possessum relation in possessive DPs. Both types of relations, I argue, come down to the abstract predicate *HOLD* (corresponding with Roversi’s (2024) *POSS*).<sup>20</sup>

- (74)a. X holds Y in its possession (possessor-possessum);  
 b. X holds Y under its control (subject-predicate).

Genitive subjects are presumably more productive in Mongolian than in other languages. This makes the language unique: it employs RX for the two kinds of reflexivity that arise between two subjects and between a subject and a possessor. Therefore, RX is involved in a single grammatical system, although it might seem to be involved in two separate systems.

This being said, we now go back to our earlier question, initially raised by Reuland (2017a): What does the specialty of reflexivity tell us about the human language system? By observing the properties of RX, a more general cognitive principle, which is expected to hold universally, is now fleshed out as follows.

#### (75) Reflexivity arises from and reflects same-subject.

One final remark in this section concerns the question, initially raised by Reuland (2017a), of why reflexivity is so special. Reuland (2017a) is devoted to answering this question. The most important one of the potentially correct answers available is probably that reflexivity is necessitated by IDI, which reflects a third factor. In other words, IDI is a necessary condition for a linguistic context in which reflexivity is realized. One way to bring IDI under the same-subject approach is to connect it with the SR-resisting property of RX in Mongolian. As noted earlier, the SR-resisting property is a property that subjects of matrix and subordinate clauses, or a local subject and a possessor must be coreferential when RX is present. That is, the sameness of subjects (or possessors as logical subjects) is indicated by RX, which is present on

<sup>18</sup> Kullmann and Tserenpil (2015: 139) call *-san* in this use “a noun determining suffix”. Janhunen (2012: 161) labels it “perfective”.

<sup>19</sup> See also Janhunen (2012: 159–160) for the nominalizing function of *-san*.

<sup>20</sup> A similar approach is den Dikken (2006), who proposes a universal functional head *RELATOR*, which mediates between any pair of elements that occupy the specifiers and complement positions.

a governing category of the lower subject (or on the possessum). In other words, RX is present to identify one subject as another (or a possessor as a subject), as required by computational systems, which are unable to identify and distinguish linguistic items with conjoint reference, without indicators such as RX and many other forms.

## 5. What is a reflexive predicate?

R&R argue that a predicate must be explicitly marked as reflexive for coreferentiality to qualify as reflexivity. Specifically, they propose the following conditions.

### (76) Conditions

A: A reflexive-marked syntactic predicate is reflexive.

B: A reflexive semantic predicate is reflexive-marked. (R&R 1993: 678)

Under their  $\lambda$ -operator approach, a reflexive predicate is a  $\lambda$ -predicate that assigns two semantic roles to its coreferential arguments. Accordingly, *love* and its arguments form a reflexive predicate in (77a) but not in (77b) and (77c), which represent what are traditionally called possessive reflexive sentences (Reuland 2017a: 26).

(77)a. Pasha<sub>i</sub> loves herself<sub>i</sub>.

b. Pasha<sub>i</sub> loves her<sub>i</sub> cat.

c. Pasha<sub>i</sub> loves the cat of herself<sub>i</sub>.

The  $\lambda$ -operator approach holds, in accordance with Condition A in (76), that when the SELF-anaphor is not a syntactic argument of the predicate, it does not have to be interpreted as a reflexivizer, but if it is, it must (Reuland 2005b: 584). Given this, *her* in (77b) and *herself* in (77c) do not make the predicates reflexive as they are not syntactic arguments of it. Nor do *her* in (77b) and *herself* in (77c) render the predicates semantically reflexive since they are not bound by the same  $\lambda$ -operator, according to (9), regardless of their reflexive interpretation. Reuland (2017a: 26–27) suggests that the bound interpretation of possessive pronouns are not syntactically encoded but represented only at the level of logical form (logical syntax in the sense of Reinhart 2006). Thus *her* in (77b) is translated as a variable in logical syntax, as illustrated below. This, however, would puzzle one as to why anaphors are not translated as variables in logical syntax.

(78) Pasha ( $\lambda x$  (loves (x [x's cat]))) (Reuland 2017a: 26)

Under the same-subject approach pursued in this paper, no such paradoxes arise because this approach does not appeal to a  $\lambda$ -operator, which requires semantic roles to be assigned to coarguments. However, for example, *her* and *Pasha* in (77b) are not coarguments, although they are coreferential. Note that *Pasha* is assigned an external role in the matrix predicate while the possessive subject, say *her*,<sup>21</sup> is assigned an external role by the semantic predicate *HOLD*,

<sup>21</sup> It must be noted that “possessive”, a semantic relation, is not “genitive”, a morpho-syntactic one. *Him* in *himself* is not genitive morpho-syntactically, but it is semantically possessor of *self* in logical syntax.

which assigns internal roles to *cat* and *self*. Thus, possessive reflexivity and anaphoric reflexivity are accounted for in a unified way, without appealing to the  $\lambda$ -operator.

One apparent merit of the  $\lambda$ -operator approach is that it is on a par with the fact that, in many languages, a possessive pronoun such as *her* does the job that would otherwise be done by a possessive anaphor, which English-type languages reportedly lack, as predicted by the Absence of Principle B effects (Rooryck & Vanden Wyngaerd 2011) and R&R's Conditions A and B.

(79) Absence of Principle B effects (APBE)

Pronouns behave like anaphors when a dedicated class of reflexive pronouns is lacking (Rooryck & Vanden Wyngaerd 2011: 19).

Unfortunately, their  $\lambda$ -operator approach does not offer a neat explanation of why some languages such as Mongolian do allow and even require possessive anaphors and why they apply a consistent reflexive marker to predicates that do not inherently involve coreferential arguments and those that do.

Therefore, the notion of reflexive semantic predicates in the sense of R&R is not universal. Reflexive semantic predicates with coreferential arguments do not exist if all anaphors are analyzable as possessive DPs (see section 6) under the same-subject approach. Nor does a reflexive syntactic predicate in the sense of R&R exist if, as assumed here, an anaphor contains a possessor, whether it is overt or implicit, which is analyzed as the external argument of the semantic predicate *HOLD* but not the internal argument of the syntactic predicate with the antecedent subject.

- (80) a. The antecedent subject is licensed by the syntactic predicate in the matrix clause.  
 b. The embedded subject is licensed by the semantic predicate *HOLD*, which is embedded under the matrix predicate.

This would lead us to the following conclusion.

(81) No reflexive predicate exists.

However, this conclusion comes too abruptly because there seems to be a consensus that examples such as (77a) as well as those with patient-reducing verbs such as *wash* count as reflexive predicates. To get the right end, let us scrutinize the structure of the anaphor under each kind of approach, taking *Pasha loves herself* as an example.

(82) Under the same-subject approach: [<sub>VP</sub> Pasha<sub>i</sub> love [<sub>DP</sub> SHE<sub>i</sub> [<sub>D'</sub> 's [<sub>N</sub> self]]]]

- a. D (=s) is a possessive determiner head in narrow syntax and a semantic predicate *HOLD* in logical syntax;  
 b. The referential part, that is, phi-features, occupy Spec-DP;  
 c. [phi] on Spec and [poss] on D (probably plus a Case feature) are spelled out as *her*;<sup>22</sup>

<sup>22</sup> See section 6 for more details.



- d. Two predicates (one being matrix and syntactic, another being embedded and semantic) are involved in the sentence;
- e. Neither of the two predicate is reflexive;
- f. Reflexivity obtains between the matrix subject *Pasha* and the logical subject *SHE*, which is contained by the DP and therefore is assigned no semantic role by the matrix predicate and is not bound by its  $\lambda$ -operator.

(83) Under the  $\lambda$ -operator approach: [<sub>VP</sub> Pasha<sub>i</sub> love [<sub>DP</sub> [<sub>D'</sub> SHE<sub>i</sub> [<sub>N</sub> self]]]]<sup>23</sup>

- a. D is a referential determiner; logical syntax is irrelevant;
- b. The referential part, that is, phi-features, occupies D;
- c. [phi] on D (probably plus a Case feature) is spelled out as *her*;
- d. Only one predicate (being syntactic and semantic) is involved in the sentence;
- e. That predicate is reflexive;
- f. Reflexivity obtains between the subject *Pasha* and *SHE-self*, which is assigned an internal role by the predicate and bound by its  $\lambda$ -operator.

(82) and (83) represent distinct derivations of *she-self* and the source of its anaphoric reflexivity. If we are to maintain the view that such sentences indeed involve reflexive predicates, we need to abandon the same-subject approach and retain the  $\lambda$ -operator approach. However, the cost of doing so is much greater than that of retaining the same-subject approach, as noted earlier and noted below. An ideal solution is to maintain the same-subject approach and find a way to reconcile it with the treatment of *Pasha loves herself* as instantiating a reflexive predicate. The crux of matter in doing so is to keep reflexivity at both levels of logical syntax and morpho-syntax. In doing so, we need to consider the possibility of an eternal combination of a pronoun such as *she* (or *her*) and *self* as a kind of lexicalization. Technically, Spec with [phi], the head D and the root N have become an inseparable morphological unit, while retaining their logical syntactic properties such as a possessor-possessum relation, which is mediated by *HOLD*. This process is expected to have been driven by a principle of economy like (84).

(84) Use as few morphemes as possible, where other principles, if any, leave the choice open.

However, the inseparability of an anaphor such as *herself* as a morphological unit does not necessarily mean that it is syntactically non-analytical. It then follows that in narrow syntax, anaphors like *herself* have the structure in (82), but not that in (83).

This entails that the representation in (82) is the first phase structure of anaphors, as is the case with verbs.<sup>24</sup> The first phase syntax indicates the internally and featurally complexity of a single lexical item.

Returning to the question of whether sentences like (77a) count as reflexive predicates, the answer will be yes, given that *herself* is decomposed into (82), with related information carried

<sup>23</sup> See R&R (1993: 658) for details of their proposal.

<sup>24</sup> Ramchand (2008: 69ff) decomposes the internal structure of *break*, for example, into a three (or four)-layered structure as shown below.

(i) Katherine broke the stick.

[<sub>init</sub>P Katherine broke [<sub>proc</sub>P the stick <break> [<sub>res</sub>P <stick><break> [<sub>XP</sub> ...]]]]



by Spec, D and N. Note that logical syntax does not interfere with the exact process of the morpho-syntactic derivation of anaphors but has an impact on their semantic interpretations, which is (partly) produced in their first phase syntax. Consequently, reflexive predicates exist but are restricted to syntactic predicates, which normally take a (non-possessive) anaphoric object. It then follows that Condition A, not Condition B, proposed by R&R remains invalid. This is an unhappy result for the  $\lambda$ -operator approach, under which Condition B universally exists.

If the  $\lambda$ -operator approach was correct with both Condition A and Condition B, another question would remain unexplained: How can reflexivity as defined by R&R be taken to be universal, whereas APBE cannot? Note that Mongolian among many other languages lacks APBE. APBE itself is not problematic. What is problematic is the definition of reflexivity in a narrow sense under the  $\lambda$ -operator approach, which only takes anaphoric reflexivity as reflexivity, excluding possessive reflexivity. Additionally, the  $\lambda$ -operator approach leaves many of the crosslinguistic facts unexplained. To mention a few: 1) Why are body anaphors and bodypart anaphors of various types attested in many languages, given that bodyparts are subject to inalienable possession? 2) Why can the pronoun contained in a SELF anaphor be genitive as in *myself*? 3) Why do languages like Mongolian take the same form to spell out an anaphoric relation and a possessive relation as well as a situational relation? Coincidence-based explanations would not be satisfactory to these questions. If possessive reflexivity is not reflexivity, all such questions remain puzzles, which, however, are all unproblematic under the same-subject approach.

It is clear that the  $\lambda$ -operator approach in fact, at least partly, appeals to syntactic argumenthood. When the bound argument bearing the internal role is overt, it must occupy a syntactic argument position, as in *John washed himself*. The motivation of incorporating semantic argumenthood into the definition of reflexivity seems to partly lie in the R&R's intention to capture lexical reflexive predicates such as *wash*, which do not take syntactic arguments if we are to maintain the idea that syntactic arguments are not omissible. R&R proposed what is called *Reduction (+Bundling)* in accounting for such predicates.

#### (85) Reduction of an internal role

$$V_{acc}(\theta_1, \theta_2) \rightarrow Rs(V)(\theta_1 - \theta_2)$$

$$V[Agent]_1 [Theme]_2 \rightarrow V[Agent-Theme]_1 \text{ (Reuland 2014: 14)}$$

However, Sportiche (2023: 21ff) provides evidence that such verbs are in fact transitives, whose the object is a silent *oneself* or *self*. Sportiche (2023: 23) adds that alternatively, when used reflexively, they involve a canonical body part and are thus cases of inalienable possession. In addition, theme-reducing verbs such as *wash* do not behave universally the same way. In Mongolian, for example, *ugaa* 'wash' requires the presence of the internal argument even when it is used in a reflexive sentence such as *Bi bey-ee ugaa-san* 'I washed my body'. Under the same-subject approach, it may be the case that the reduction of the internal role does not take place; it does not undergo bundling with the external role. The absence of the internal argument for such verbs is not a reduction of the internal role, but rather a lack of morpho-phonological content of the whole DP, which would otherwise bear the internal role. The whole DP can be absent due to the minimal-nounhood of the "possessum" in the possessive DP as well as

language-specific morpho-syntactic properties.<sup>25</sup>

(86) [DP possessor [D' D [ N/R(=possessum) ]]]

Recall that the full structure of anaphors involves three elements, namely the possessor, the possessive D head and the possessed N. Any of these elements may lack morpho-phonological content, as required by an economy principle as informally stated in (84), which requires a smallest, in number and length of the syllables, morpho-phonological unit to spell out the elements and may even require nothing if the interpretation goes right.

It is expected that there will be no independent universal constraint other than (84) on how the elements reach the morphological component. That is, language-specific morpho-syntactic rules are at play. Ultimately, the so-called reduction of the internal role of *wash* verbs turns out to be a parametric matter of morpho-syntax, with a semantic effect, but not simply a matter of semantics.

In summary, so-called reflexive predicates, if there is a need of characterizing them, are syntactic predicates that assign semantic roles to their coarguments. This is basically the same stance as Condition A. In this characterization, the whole DP structure of the anaphor comes with an effect of “squeezing out” the Spec-D-N relation and a predication interpretation, where, however, the semantic predicate *HOLD* is still at play. Note that logical syntax does not interfere with the exact derivation of anaphors. Therefore, it can be said that the so-called “reflexive predicate” does not exist in the strict sense; what exists is a property of being “used-as-a-reflexive-predicate”.

## 6. What is an anaphor?

As described by many previous studies, anaphors are formed by means of a syntactic operation such as binding, although studies such as Reuland (2020a: 3) hold that syntactic binding does not give rise to the syntactic category “anaphor”. The same-subject approach perused in this paper, on the basis of cross-linguistic data, argues for a view that anaphors are formed by clustering features and a minimal noun, and form a homogenous class. Chinese, Japanese, English and Dutch are discussed. Assuming that (86) as the full structure of anaphors holds universally, I present the following proposals and arguments for them.

(87) Nature of anaphors and pronouns

- a. A simplex anaphor is a bundle of a possessive feature, a reflexive feature and a nominal root with minimal lexical content.
- b. An anaphor contrasts a pronoun, which is a bundle of phi-features, lacking what a anaphor has.
- c. A complex anaphor is thus a bundle of what a simplex anaphor and a pronoun have.

*Zi-* in Chinese *zi-ji*, which is a simplex anaphor, is a morpheme that contributes to a reflexive interpretation (Reuland et al. 2020). The second part *-ji* behaves like a minimal noun or a

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<sup>25</sup> If we want to maintain that there is indeed reduction of internal roles, leading to lexical reflexivity of verbs such as *wash*, under the same-subject approach, it may be the case that object anaphors (of the predicate) with a first phase syntax are incorporated into the verb in one way or another.

nominal root. Interestingly, *zi-* can also take *-shen*, which literally means “body” (Inoue 2014: 23) to form an alternative anaphoric form *zi-shen* (Wang and Pan 2021: 16). Arguably, this also holds for *zi-ge’er*, in which *-ge* means “body” or “body size”. *Zi-shen* is always used adnominally and *zi-ge’er* is used adverbially, unlike *zi-ji*, which functions as both an adnominal and an adverbial in addition to arguments.

(87) Lisi    zai    zebei    zi-ji.

Lisi    ASP    blame    self

Lisi is blaming himself. (adapted from Tang 1989: 94)

(88) Ta    budebu    zhengshi    zi-shen    zuowei    fuqin    de    buzu.

he    have to    face up    self    as    father    GEN    inadequacy

‘He had to face up to his own inadequacies as a father.’

(Oxford Advanced Learner's English-Chinese Dictionary)

The following structure represents the formation of *zi-* anaphors in Chinese.

(89) [DP *zi-* [D' D [ -ji/shen/ge'er ]]]

Importantly, a pronoun such as *ta* ‘(s)he’ can select *zi-ji* to form a complex anaphor *ta zi-ji* ‘himself/herself’. Given that morpho-syntactically, a pronoun is just a bundle of phi-features and lacks lexical content (Reuland 2017a: 48), the following structure obtains.

(90) [DP D<sub>[phi]</sub> D<sub>[ref]</sub> [D' D<sub>[poss]</sub> [ N ]]] (merger of (D heads with) features)

(91) [DP *ta* *zi-* [D' Ø [ -ji/shen/ge'er ]]] (lexical insertion)

The morphemes *ta*, *zi-*, and *-ji* spell out [phi], [ref] (for reflexivity), and N, respectively, with [poss] (for possessivity) lacking a morphological realization. [phi], [ref] and [poss] are located on three bare heads,<sup>26</sup> where D<sub>[phi]</sub> and D<sub>[ref]</sub> are present as stacked specifiers of D<sub>[poss]</sub>, which is the head of the phrase. This analysis is based on Davis (2023).

In Japanese anaphor *zi-sin*, *zi-* contributes a reflexive interpretation, as is the case with Chinese *zi-*, and *-sin* contributes a lexical interpretation as “body”, as is the case with Chinese *-shen*. The anaphor *mi*, which stands alone, literally means “body” (Nishida 2002: 272; Noguchi 2018: 2). Japanese has another *zi-* anaphor, namely, *zi-bun*, in which the prototypical meaning of *-bun* is “portion” or “part”, and this meaning is retained in the anaphor to some extent.

(92) Taroo-ga    zi-bun-o    semeta.

Taro-Nom    self-Acc    blamed

‘Taro criticized himself.’ (Tsujimura 2014: 255)

<sup>26</sup> In line with the theory of Bare Phrase Structure (Chomsky 1995), Davis (2023) proposed that English genitive pronouns are created from features on D heads, which have an in-between status, without projecting phrases but occupying the specifier of another D head.

- (93) John-ga kare zi-sin-o hihansi-ta.  
 John-NOM himself-ACC criticized  
 ‘John criticized himself.’ (Nakamura 1989: 207)

Similarly, the pronouns *kare* ‘he’ and *kanozyo* ‘she’ can select *zi-sin* to form complex anaphors, whose derivation is represented below.

- (94) [DP D<sub>[phi]</sub> D<sub>[ref]</sub> [D' D<sub>[poss]</sub> [N ]]] (merger of (D heads with) features)

- (95) [DP kare zi- [D' Ø [ -sin ]]] (lexical insertion)

If this structure under the same-subject approach is on the right track, it has to explain non-possessive anaphors including SELF anaphors such as English *himself* and SE anaphors such as Dutch *zich*. Conceptually, SELF may correspond to the minimal noun N, which is “possessed” by the pronoun selecting it, say, *him*. Specifically, the entity with [3rd, m, sg], which is coreferential with a local subject, holds SELF. In this sense, *himself* is to be conceptually construed as *his-self*, with a possessive relation involved, as is the case with Mongolian *öör-in bey* ‘own body’ and bodypart reflexives attested in many languages. With respect to why the pronoun in *himself* is the accusative *him* rather than the possessive *his*, a morpho-syntactic parameter is arguably at play. This is even more reasonable given that there are *myself* and *yourself* as well as *herself* in English.

- (96) [DP D<sub>[phi]</sub> D<sub>[ref]</sub> [D' D<sub>[poss]</sub> [N ]]] (merger of (D heads with) features)

- (97) [DP D<sub>[phi, poss]</sub> [DP D<sub>[ref, poss]</sub> [N ]]] (clustering of features = fusion)

- (98) [DP my [D' Ø [ self ]]] (lexical insertion)

As illustrated in (119–120), the features that enter the derivation separately undergo fusion, followed by application of the vocabulary insertion rule, which chooses the morpheme *my* for the fused outcome D<sub>[phi, poss]</sub>, as argued by Davis (2023) and Bai et al. (2025). *Self* is chosen for N. D<sub>[ref, poss]</sub> is not spelled out overtly. Alternatively, D<sub>[ref, poss]</sub> or D<sub>[ref]</sub> is spelled out by *self*, with N lacking a realization.

In this sense, the semantic predicate *HOLD* is present in English anaphors such as *myself*, which can be paraphrased by “I hold SELF”. The difficulty of speakers’ interpreting *self* as a possessum arguably lies in the fact that it is highly grammaticalized, having lost its intensifying function to a great extent. Yet *self* is not entirely empty in meaning. Its occurrence in anaphors is still at play in logical semantics, going with an “the subject holds SELF” effect.

Thus, in, for example, *picture of oneself*, stacked possession is involved, as is the case with Mongolian *öör-in bey-in-ee čadal* ‘own/self’s body’s strength’. Note that the pronoun in English anaphors is not omissible arguably because English is not a zero-determiner language. In contrast, *öör-in* in Mongolian, which is a zero-determiner language, is always omissible, and the presence of a pronoun in anaphors is not preferable or even disallowed (see section 2).

This allows us to say that SE anaphors are in fact a shortened form of SELF anaphors. For

example, the Dutch *zich* spells out the possessor occupying the specifier of D that selects N. N is optionally spelled out by *zelf*, as in *zichzelf*.

(99) [DP D<sub>[phi]</sub> D<sub>[ref]</sub> [D' D<sub>[poss]</sub> [N ]]] (merger of (D heads with) features)

(100) [DP Ø zich [D' Ø [Ø ]]] (lexical insertion)

However, the morpho-syntactic behaviors of SELF anaphors and SE anaphors in different languages are quite arbitrary. One language may allow only one of a SELF anaphor and a SE anaphor while another language may allow both, and one language may allow a complementary distribution while another does not. The factors that determine the ultimate shape and behaviors of anaphors are by no means restricted to semantic factors, although reflexivity, a semantic notion, is universal. The exact shape of an anaphor is determined by language specific spell-out rules. It is thus reasonable to say that ways of encoding reflexivity are sensitive to a morpho-syntactic parameter and syntactic operations that drive relevant features to get clustered determine the syntactic binding of the possessor, as partially indicated by Reuland's (2014: 21) statement that standard morpho-syntactic features such as specification for phi-features, Case, denoting a relation, and so on together with general principles of derivation determine whether an element will be bound.

Among the four languages we have discussed, English is special in that it disallows [phi] to remain shapeless. However, it allows N or [ref] lacking a morphological realization. This is arguably because in English, [poss], not [ref], is obligatorily clustered with [phi], as discussed by Davis (2023) in detail. Japanese differs from others in that [phi] is obligatorily spelled out as *kare* or *kanozyo*, which precedes *zi-sin*. On the other hand, [phi] is not spelled out in the case of *zi-bun*. In Chinese, [phi] is optionally spelled out as *ta*. Dutch is closest to Mongolian in that either [ref] or [ref] plus N is spelled out: *zich* and *zichzelf* in Dutch and *öör* and *bey*. However, Mongolian differs from all others in that [poss] is obligatorily clustered with [ref], with the clustered outcome [ref, poss] spelled out as *öör-in*. Note that clustering of features, or fusion of relevant D heads, is not arbitrary. There are certain constraints, as discussed by another paper in preparation by the author.

Note that in all the cases discussed above, [phi] enters the derivation unvalued and gets valued by Agree with [phi] on the subject, instantiating what we call binding (Reuland 2014; Rooryck and Vanden Wyngaerd 2011). In this sense, coreferentiality is obtained between the possessor with [phi] and the subject. Which one of [ref] and N surfaces as a marker of the coreferentiality or a reflexivizer depends on specific morpho-syntactic properties of individual languages.

## 7. Summary and discussion

According to Cohen and Zribi-Hertz (2014), scholarship on reflexivity is divided into two primary camps: those viewing it strictly as an anaphoric relation, and those examining it as one of several semantic outcomes triggered by reflexive morphology. This paper argued that the latter perspective represents the most vital direction for future inquiry. To advance this field, this paper investigated Mongolian reflexive strategies, illustrating how reflexivity interacts with varied semantic effects and advocating for a more comprehensive definition of the phenomenon.

The specific findings are detailed below. First, reflexivity is a referential identity between subjects, arising in need of remedying the effect of IDI, where the antecedent subject is of a syntactic predicate and the other one is of another syntactic predicate or of the semantic predicate *HOLD*. This remains a more general principle, which holds universally. Speaking in terms of the X-bar schema, reflexivity arises between specifiers, not between a specifier and a complement. Second, reflexive predicates do not exist in the strict sense but a property of “used-as-a-reflexive-predicate” may exist. Third, a simplex anaphor is a morphological realization of a bundle of possessive and reflexive features with a minimal noun; a complex anaphor is formed by the combination of a simplex anaphor and a pronoun. Fourth, possessivity (or possessor-possessum relation) is a subcase of predication (or subject-predicate relation) by nature in that it is established on the basis of the abstract predicate *HOLD*, which licenses subjecthood to possessors.

Below is a brief discussion of what can be implied by above conclusions. In the Mandarin sentence (101), the reflexivizer *zi* ‘self’ is not an independent nominal but an affixal element attached to the verb *ze* ‘blame’. The Mongolian counterpart of this sentence must use RX in addition to an anaphor.

- (101) *Lisi zi-ze.*  
*Lisi self-blame*  
 ‘Lisi blames himself.’

In (102), *home* always refers to John’s own home. The Mongolian counterpart of this sentence must use RX, not the anaphor, to indicate that the home is the subject’s home.

- (102) *John went home.*

In (103), it is often assumed that the predicate is lexically reflexive, with the internal role reduced into it. Again, the Mongolian counterpart of this sentence must use RX in addition to an anaphor.

- (103) *John washed.*

Reflexivity is thus involved in all these sentences, and it is additionally marked in their Mongolian counterparts. Accordingly, reflexivity is not necessarily anaphoric-marked or even zero-marked cross-linguistically. An anaphoric element used to mark reflexivity may be either an independent nominal or a lexical part of the predicate. All this indicates that the mechanism, if any, connecting so-called reflexive predicates and reflexivity marking is not universal, contrary to what is implied by R&R’s claim (11). Instead, ways to mark reflexivity vary widely across languages, where the morpho-syntactic properties of the determiner system, the possessive structure, verb-formation rules, and so on in individual languages play important roles in deriving the parametric variation of reflexivity marking.

With the simultaneous existence of sentences such as (102) and (103) indicates that reflexivity can also be lexically encoded. In that case, reflexivity does not arise from the identity between elements in specifier positions. This requires a theoretical boundary between



syntactically conditioned reflexivity and lexical reflexivity. This may suggest that documented instances of reflexivity are situated along a “reflexivity continuum”. Within this spectrum, same-subject reflexivity and lexical reflexivity (as in *wash* verbs) represent the two extremes of the continuum. If this is the case, reflexivity turns out to be not conditioned by a universal manner. Consequently, it appears that neither the same-subject approach explored in this paper nor R&R’s subject-object approach (the  $\lambda$ -operator approach) is entirely definitive. Instead, each theory may uniquely account for a distinct manifestation of reflexivity. Although a more exhaustive investigation of this issue lies beyond the scope of this study, the proposed same-subject approach provides a more accurate representation of reflexivity, having made it more approximate to what it is. By moving toward a broader generalization—specifically for syntactically-conditioned reflexivity—this approach offers a more comprehensive theoretical account.

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